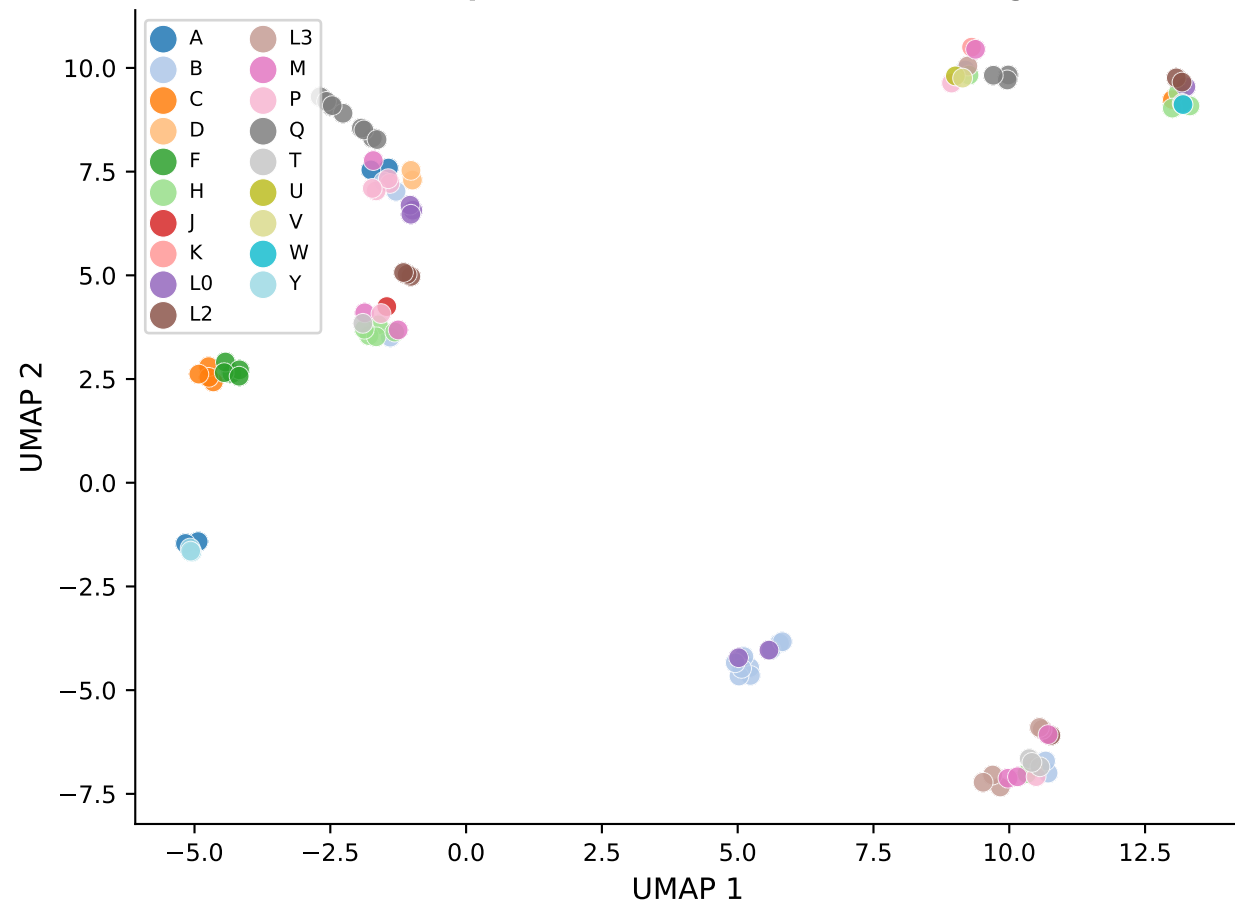


Figure 3: Haplogroup embedding structure and 26-class zero-shot classification

**(a) mtDNA-FM embeddings in UMAP space
(100 sequences, zero-shot, no fine-tuning)**



**(b) Zero-shot haplogroup classification
(26-class panel, 5-NN cosine)**

